

The Workflow Is Not Enough

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Abstract. Purposively designing organizations to employ non-human knowledge workers is essential to generating stronger performance and goal attainment. Recent studies suggest that workflow configurations can encourage human knowledge workers to embrace and work more collaboratively with non-human knowledge workers, especially configurations that strengthen role clarity and autonomy for human group members. However, results from virtual experiments indicate a more systematic approach is needed—an approach that adequately considers both agential and organizational factors—otherwise desired gains in group performance might not be realized. This study explores the simulated effects on organizational performance of six workflow configurations that emerged from recent studies on collaborative decision-making by human and synthetic computational agents. This paper recommends future studies emphasize field experimentation to corroborate intellectual, agent-based, social network, and organizational engineering models of existing and emerging organizations and to strengthen experiment validity.

Keywords: Human-AI Teaming, Organizational Design, Group Performance

1 Introduction

Organizations are increasingly incorporating synthetic computational agents into their operations to strengthen their performance and generate greater returns. Organizational performance depends largely on how group members transform system-level inputs into outputs; group members apply their knowledge and skills using relevant techniques and tools to accomplish tasks [1, 2]. Organizational performance, therefore, largely depends on the abilities possessed by group members. Cognitive, physiological, and similar factors can limit human and, consequently, organizational performance [3, 4]. Modern synthetic computational agents possess capabilities that can exceed those of humans, leading organizations to expand their use to strengthen performance and goal attainment [5, 6]. The extent to which synthetic computational agents contribute to organizational performance and goal attainment, such as greater revenue generation, probably depends on how comprehensively group designs consider both agential and system-level factors [5, 7].

Intentionally designing organizations to employ synthetic computational agents more optimally is crucial. Organizational design entails engineering how group members transform system inputs into outputs [1, 2]. In other words, organizational design

effectively determines how an organization converts orders and invoices into products and services. The approach to incorporating synthetic computational agents is important; roughly 30% of an organization’s performance results from how well its design fits its environment [8, 9].

Workflow configuration can assist with more gainful employment of non-human knowledge workers (NHKWs). NHKWs are synthetic computational agents primarily characterized by computational power commensurate with some human knowledge workers (HKWs), mostly knowledge work assignments, task-level performance, and more comprehensive incorporation into organizational networks [10]. However, virtual experiments indicate that workflow configuration alone generally does not result in meaningful gains in group performance. This paper begins with an overview of organizational systems and then describes the methodology used in virtual experiments. Experiment results are summarized and discussed next, along with limitations and recommendations for future studies.

2 Organizational Systems

Organizations are purposively designed systems that enable group members to attain shared goals and overcome limitations on individual performance. An information-processing and communication abstraction is commonly used to model organizations and task performance [11–14]. Limitations exist on the volume of information that individual group members can locate, gather, process, curate, and share [15–18]. Divisions of labor, cognitive and manual, enable group members to benefit from expertise, information, and other resources distributed throughout the system, strengthening organizational performance and goal attainment [4, 19, 20]. Divisions of labor necessitate configuring and coordinating work between group members, making organizational performance more dependent on interdependently performed work [1].

Organizations are also complex systems. An organization consists of a set of interdependent group members, who contribute to and benefit from the system through a non-standard set of communication paths and mechanisms [20–22]. As a complex system, an organization is also interdependent with its environment. Complexity results from the non-standard set of communication paths and mechanisms, and emergent behaviors that are irreducible to individual constituents.

2.1 Organizational Representations

Organizations comprise an interdependent set of task, task coordination, and communication networks through which group members cooperate and coordinate the use of resources to accomplish work [14, 23, 24]. Task networks represent how group members perform work. Task networks emulate major groupings of activities, the sequence and interrelationships of such groupings, needed expertise and skills, and similar factors [14, 25, 26].

Task coordination networks generally emulate organizational reporting hierarchies [14, 25, 26]. Task coordination networks represent group members, their roles and

interrelationships, possessed expertise and skill levels, and availability. Simply, task coordination networks describe who works for whom, and their knowledge and skills.

Communication networks normally represent information sharing paths and mechanisms by which group members coordinate task performance [14, 25, 26]. Communication networks include meetings, information stores, information sharing mechanisms, and similar means by which group members locate and exchange information. Organizational engineering models are characterized by their formal representations of task, task coordination, and communication networks [27].

2.2 Workflow Configurations

Task sequencing and differentiated expertise are two parameters that often characterize workflow configuration. Task sequencing describes the order in which group members perform distinct tasks [20, 28, 29]. Group members can perform tasks either serially or in parallel. In a serial design, Task A is performed before Task B and the outputs of Task A frequently serve as inputs to Task B. In a parallel design, Tasks A and B are performed concurrently, overall. Divisions of labor mean group members must finish both tasks for the organization to complete work.

Differentiated expertise describes group members possessing generally unique cognitive specializations. An organization characterized by non-specialized knowledge is one in which different group members generally know the same things, i.e., an integrated transactive memory system [28–30]. In contrast, an organization characterized by specialized knowledge is one in which different group members generally know different things, i.e., a differentiated transactive memory system.

Permutations of task sequencing and expertise generated the six workflow configurations depicted in Fig. 1. Configurations 1, 3, and 4 illustrate parallel and serial configurations for an organization with non-specialized knowledge. Configurations 2, 5, and 6 illustrate parallel and serial configurations for an organization with specialized knowledge. The parallel task sequencing in Configurations 1 and 2 results in a single design for each organization with non-specialized and specialized knowledge, respectively. These six workflow configurations emerged from analysis of qualitative data collected from nearly 260 HKWs in four countries regarding their perceptions of NHKWs and collaborative decision-making [6, 28, 29].

3 Methodology

Workflow configurations, alone, might not be enough to significantly strengthen organizational performance. The six configurations depicted in Fig. 1 resulted from survey data collected by [6]. Respondents generally indicated that Configuration 2, parallel task sequence with specialized expertise, provided greater role clarity for HKWs, contributing to greater confidence in NHKWs and group performance. Configuration 3, serial task performance with a NHKW performing work before a HKW and without specialized expertise, can seemingly reduce HKW confidence in NHKWs

and their performance. This lesser degree of confidence appears to stem from HKWs performing and scrutinizing the same work NHKWs accomplished.

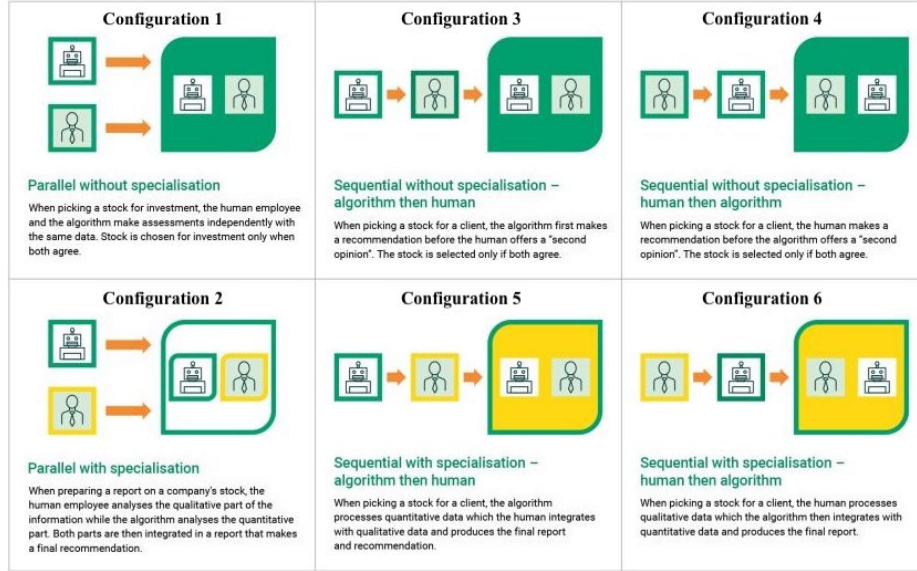


Fig. 1. Workflow configurations representing task sequence and differentiated expertise permutations. Adapted from [29].

Organizations provide members with the means to attain shared goals, such as stronger group performance and returns on investments. Therefore, the impacts of HKW-NHKW workflow configurations on organizational performance are a crucial design consideration. Although respondents provided valuable insights, empirical exploration of the effects the six configurations might have on organizational performance is largely absent [6, 28, 29]. Virtual experiments in this study address that gap.

3.1 Virtual Experiments

Computational organization theory (COT) provides a robust framework for virtual experiments, supplementing laboratory and field studies. Virtual experiments enable researchers to generate, operationalize, and test theories [14, 31–33]. Many COT models, including agent-based, social network, and organizational engineering models, result from qualitative studies [8, 9, 14, 27, 34], such as [6].

The validity of virtual experiments largely depends on three considerations. The purpose of the virtual experiment is first [33, 35]. The extent to which specific computational model features fit experiment purposes is second [33, 35]. Experiment design, including data analysis techniques, is third [33, 35]. When balanced, “the purpose remains the anchor” of these three considerations [33, p. 65].

3.2 Computational Modeling Application

This study used Project Organization and Workflow for Edge Research (POWer) to conduct virtual experiments. POWer is an organizational engineering COT modeling tool, intentionally designed to study knowledge work-centric organizations [14, 36]. A relevant strength of organizational engineering models is their formal, comprehensive representations of task, task coordination, and communication networks [27], such as the workflow configurations described by [6, 28, 29].

POWer is designed to emulate knowledge work and treats the organization as the primary unit of analysis, further strengthening its relevance for virtual experiments exploring the six workflow configurations. Knowledge work can involve specialized and non-specialized forms of expertise [28, 30, 37], as well as varying skill levels between group members [14, 37]. POWer provides the ability to manipulate both the expertise areas and skill levels possessed by group members. Essential to this study is exploring impacts each of the six workflow configurations are likely to have on organizational performance. By treating the organization as the unit of analysis, instead of individual group members, POWer can provide important insights on group performance effects absent from the qualitative data collected [6, 28, 29].

3.3 Experiment Scenarios and Design

Picking an investment stock and preparing a report on the company's stock served as the scenarios used to model the six workflow configurations. Using data from the study reported by [6], [28, 29] developed the six workflow configurations illustrated in Fig. 1. Picking an investment stock provided the context for five of the six workflow configurations [29], Configurations 1, 3, 4, 5, and 6. In the stock picking scenario, the HKW and NHKW must agree on the stock picked, regardless of task sequencing and expertise. Meanwhile, preparing a report on the company's stock performance served as the scenario for Configuration 2, which represents parallel task sequencing with specialized expertise [29]. In this scenario, the HKW and NHKW possess specialized expertise in qualitative and quantitative data analysis, respectively.

Virtual experiments modeled the two scenarios with baselines consisting of two HKWs and treatments consisting of one HKW and NHKW. Virtual experiments followed the design outlined by [26], using a 5% significance level. For each experiment:

- The probability a group member makes an error resulting in rework was 10%.
- The probability a group member needs to communicate with the other group member was 5%.
- "Stock Picking" served as the common task for Configurations 1, 3, and 4.
- "Qualitative Data Analysis" served as the HKW expertise area and "Quantitative Data Analysis" served as the NHKW expertise area for Configurations 2, 5, and 6.
- HKWs and NHKWs possessed "medium" level skill proficiency for their respective expertise areas. It took one-tenth of a workday (roughly 48 minutes) for group members to initially develop or regain proficiency in their assigned expertise areas.

- Tasks took two workdays to perform, notionally.
- Group members worked the same eight hours each day, five days per week.
- HKWs possessed the capacity of one full-time equivalent (FTE) employee and fully dedicated to their assigned tasks, without any external disruptions. NHKWs possessed capacity equal to ten FTEs to emulate nearly unlimited information-processing capacity and, likewise, were fully dedicated to their assigned tasks, without any external disruptions.

The difference in mean *total work volumes* (TWVs) served as the criterion in the virtual experiments. TWV is the sum of the production, rework, coordination, and waiting time work volumes and represents the cumulative work an organization performs [14, 25, 26]. Production work volume refers to cognitive and other labor performed to accomplish assigned tasks. Rework volume describes the amount of work performed to address either mistakes that they or other group members made. Coordination work volume represents the amount of work needed to share relevant expertise and information. Finally, waiting time work volumes describe the impacts incurred from waiting for other group members to complete supporting work.

3.4 Hypotheses

It is hypothesized that simulated mean TWVs for:

- Configurations 1 and 2, and their respective baseline configurations are equivalent;
- Configurations 3, 4, 5, and 6 are less than their respective baseline configurations;
- Configurations 3 and 4 are equivalent; and
- Configurations 5 and 6 are equivalent.

4 Results

Results from virtual experiments are largely consistent with anticipated outcomes and are summarized in Fig. 2. Virtual experiments used data generated from three simulations ($n = 1,000$ trials per simulation) for each of the 12 workflow configurations – six baseline and six treatment configurations. Statistical tests for the differences in mean TWVs were conducted using the generated data at a 5% significance level.

Six of the eight experiments generated results consistent with expectations. Statistical analyses, using a two-tailed z-test, identified no meaningful differences between mean TWVs for Configuration 1 and its baseline, and Configuration 2 and its baseline. In short, neither Configuration 1 nor Configuration 2 generated stronger organizational performance, regardless of whether the workflow configuration considers non-specialized and specialized expertise. Similarly, statistical analyses, again using a two-tailed z-test, indicated no meaningful differences between mean TWVs for Configurations 3 and 4, and Configurations 5 and 6. Swapping the order in which an HKW and NHKW performed work did not generate stronger organizational performance, regardless of non-specialized or specialized expertise, in serial task sequence designs. Confidence interval analyses corroborated z-test results.

Simulated Baseline Total Work Volumes (TWVs; in days)											
Baseline Configuration 1			Baseline Configuration 2			Baseline Configuration 3			Baseline Configuration 4		
TWV	TWV Standard Deviation		TWV	TWV Standard Deviation		TWV	TWV Standard Deviation		TWV	TWV Standard Deviation	
Simulation 1	4.15	0.39	4.15	0.40		4.22	0.43		4.22	0.44	
Simulation 2	4.15	0.38	4.14	0.38		4.22	0.44		4.21	0.41	
Simulation 3	4.15	0.37	4.14	0.36		4.23	0.46		4.20	0.41	
Mean values	4.15	0.38	4.14	0.38		4.22	0.45		4.21	0.42	
Simulated Treatment TWVs (in days)											
Configuration 1			Configuration 2			Configuration 3			Configuration 4		
TWV	TWV Standard Deviation		TWV	TWV Standard Deviation		TWV	TWV Standard Deviation		TWV	TWV Standard Deviation	
Simulation 1	4.25	0.60	4.21	0.53		4.20	0.51		4.06	0.23	
Simulation 2	4.23	0.57	4.20	0.54		4.17	0.50		4.08	0.28	
Simulation 3	4.22	0.56	4.22	0.54		4.22	0.56		4.06	0.23	
Mean values	4.23	0.58	4.21	0.54		4.20	0.52		4.07	0.25	
Experiment Hypotheses		z-Tests			Confidence Interval Analysis (in days)						
		z-Critical	z-Statistic	Result	Difference in Means	Lower Bound	Upper Bound	Result			
1. Baseline Configuration 1 and Configuration 1 TWVs are not equivalent		+/-1.96	-3.635	Reject	0.08	0.037	0.123	Corroborates			
2. Baseline Configuration 2 and Configuration 2 TWVs are not equivalent		+/-1.96	-3.212	Reject	0.07	0.026	0.100	Corroborates			
3. Configuration 3 TWV $\geq$ Baseline Configuration 3 TWV		1.645	1.194	Fail to reject	-0.03	-0.069	0.017	Indeterminate			
4. Configuration 4 TWV $\geq$ Baseline Configuration 4 TWV		1.645	9.258	Reject	-0.14	-0.200	-0.113	Corroborates			
5. Configuration 5 TWV $\geq$ Baseline Configuration 5 TWV		1.645	1.639	Fail to reject	-0.04	-0.079	0.007	Indeterminate			
6. Configuration 6 TWV $\geq$ Baseline Configuration 6 TWV		1.645	10.536	Reject	-0.15	-0.179	-0.123	Corroborates			
7. Configuration 3 and Configuration 4 TWVs are not equivalent		+/-1.96	6.987	Reject	0.13	0.092	0.164	Corroborates			
8. Configuration 5 and Configuration 6 TWVs are not equivalent		+/-1.96	6.590	Reject	0.13	0.091	0.168	Corroborates			

Fig. 2. Summary of virtual experiment statistics, at a 5% significance level ($n = 1,000$ trials per simulation). Results for Configurations 3, 5 and their associated baselines were indeterminate, according to statistical analyses performed, and are highlighted in yellow.

Configurations 4 and 6 generated stronger organizational performance relative to their respective baselines, as anticipated. Statistical analyses, using one-tailed z -tests, indicated that substituting an NHKW for a HKW in serial task sequence designs generated stronger organizational performance for both non-specialized and specialized expertise cases. Again, confidence interval analyses corroborated z -test results.

Experiment results for the remaining two workflow configurations, Configurations 3 and 5, were indeterminate. Statistical analyses, using one-tailed z -tests, indicated that no meaningful differences existed between mean TWVs for Configurations 3 and 5 and their respective baselines, contrary to anticipated outcomes. However, confidence interval analyses failed to corroborate the z -test results for these two configurations. Therefore, these two experiments generated indeterminate results. Although Configurations 3 and 5 represent serial task sequencing designs in which an NHKW performs work before an HKW, intuition suggests the indeterminate findings are more likely due to the stochastic nature of the virtual experiments than task sequence.

5 Discussion

Researchers and managers need to balance approaches likely to strengthen HKW acceptance of NHKWs and group performance in organizational designs. Stronger role clarity and employee autonomy contribute to greater confidence in NHKWs, resulting in respondents generally preferring Configuration 2 over the other five configurations [6, 29]. However, stronger role clarity and autonomy do not necessarily translate to more robust organizational performance, as virtual experiment results indicated. This is a crucial consideration for organizational designers and managers. Organizations are increasingly investing in and incorporating NHKWs and other forms of synthetic computational agents into their operations [5, 6]; however, organi-

zational returns on investment and performance might suffer needlessly, unless adequate consideration is given to agent-level and system-level impacts on performance.

Two limitations associated with this study are significant. First, the workflow configuration and virtual experiment models are relatively simple, lacking the degree of complexity inherent to modern organizations, overall. The intended purpose of a COT model serves as the primary measure of its validity, and unnecessarily complex models can diminish the value of virtual experiments [34, 35]. Additionally, micro-theory suggests that exploring HKW and NHKW micro-behaviors, even in relatively small organizations, can yield valuable insights about system-level performance [14]. Thus, the relative simplicity of the workflow configuration and virtual experiment models can be considered a strength, especially at this stage of exploration.

Second, POWER assumes that HKWs and NHKWs are fully productive, except for disruptions from other individuals. More simply, POWER does not consider productivity variations over the course of a workday and associated effects on organizational performance. Nevertheless, empirical studies used to develop and test POWER and its predecessor indicate this modeling simplification is reasonable [12, 25, 26].

Designing modern organizations for more optimal performance necessitates expanding our understanding of how NHKWs are likely to affect the performance of existing organizations, which are largely human-centric. Future studies should include field experimentation to corroborate intellectual, agent-based, social network, and organizational engineering models of existing and emerging organizations. Additionally, future studies should address anomalies in this and other relevant studies, such as the non-determinate findings from experiments exploring the impacts of Configurations 3 and 5 on organizational performance.

6 Conclusion

A systematic approach to balancing agential and organizational considerations seems crucial to designing organizations comprising HKWs and NHKWs for stronger performance and goal attainment. HKW confidence in and willingness to work alongside NHKWs likely shape the degree to which organizations reap benefits from investing in synthetic computational agents. Previous studies indicate that organizational workflow configurations can encourage HKWs to embrace NHKWs more fully, especially configurations that provide greater role clarity and autonomy. However, virtual experiments indicate that organizational designers and managers must consider system-level factors, such as the effects that coordination and other emergent behaviors have on group performance, if NHKWs are to contribute more impactfully to organizational performance and goal attainment. Importantly, NHKWs are present in some organizations already, and both the number of NHKWs and scope of their roles are likely to expand, making further exploration of HKW-NHKW organizational designs an imperative for COT community members.

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